Tidal potential measured on a coastal–inland optical-clock baseline

Abstract

The Moon and the Sun alter Earth's gravitational potential through their tidal fields and the responses of the solid Earth and oceans. Changes in the tidal potential modify the relative ticking rates of clocks at different locations. Advances in optical clocks and fiber-optic network technologies have made clocks precise probes for probing spacetime, providing an experimental foundation for the long-term vision of monitoring Earth's gravitational potential with a global clock network. In this work, we use strontium and ytterbium clocks separated by 692 km to detect tidal variations in the potential difference between the coastal region of Shanghai and the inland region of Wuhan. Based on 545.9 hours of observations spanning 353 days, the joint measurement of solid-Earth and ocean loading agrees with model predictions with a consistency of 0.96 ± 0.11. Using spectral analysis, we separated the principal lunar semidiurnal tidal constituent M_2 and the principal solar semidiurnal tidal constituent S_2, with frequencies of 22.361 μHz and 23.147 μHz, respectively, and signal-to-noise ratios of 16.7 dB and 9.3 dB, respectively. This experiment demonstrates that optical clock networks can already be used for high-precision gravitational potential calibration and detection, laying a foundation for related applications.